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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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1. Introduction

This project asks whether machine learning tools improve portfolio construction or whether they add complexity without adding value. We start with a Kelly-optimal portfolio of ten large US stocks, then try three covariance-matrix techniques, including denoising, hierarchical clustering, and detoning, and then test whether any of them earns its keep.

Our own results give a qualified answer. One technique helped consistently, one made no measurable difference, and the simplest benchmark of all stayed competitive with everything else. We report the outcome exactly as we found it.

1.1 Data and windows

The same ten securities carried over from GWP1: TSLA, WMT, BAC, GS, LLY, MRK, GOOG, META, AAPL, and XOM. We work with daily adjusted closes covering 502 trading days. The training window runs from 3 January 2024 to 30 September 2025 (437 observations); the test window runs from 1 October to 31 December 2025 (64 observations). The two windows do not overlap. Step 3 uses the full set of 501 return observations. We take the risk-free rate of 4.50% per year.

Table 1: In-sample descriptive statistics

Ticker	Ann. mean	Ann. volatility	Skew	Excess kurtosis
TSLA	54.74%	65.59%	0.67	4.51
WMT	41.68%	21.49%	0.72	9.08
BAC	29.94%	25.40%	−0.64	8.78
GS	47.91%	29.09%	0.69	11.18
LLY	21.96%	36.32%	−0.35	8.43
MRK	−10.84%	25.32%	−0.88	6.89
GOOG	36.99%	29.98%	0.25	5.15
META	50.68%	37.75%	1.74	16.33
AAPL	22.90%	29.13%	1.00	13.19
XOM	11.49%	21.89%	−0.62	2.44

437 daily observations. Every asset shows excess kurtosis, ranging from 2.4 for Exxon to 16.3 for Meta. Returns are far from normal. That matters because Kelly maximizes log growth, which is sensitive to the left tail. We therefore optimize directly on the empirical return distribution rather than a Gaussian approximation.

1.2 A note on the brief

Step 2 asks us to compare “the Kelly portfolio with the double-Kelly and double-Kelly portfolio.” We read this as the standard fractional comparison: half-Kelly, full-Kelly, and double-Kelly, which is the only interpretation that produces three distinct portfolios.

2. Step 1: The Optimal Kelly Portfolio

2.1 What Kelly maximizes

The Kelly criterion picks the portfolio with the highest expected log wealth, which is the same as the highest long-run geometric growth rate (Kelly 1956):

$$\max g(w) = E[\ln(1 + w^T r)] \text{ subject to } \sum w_i = 1, 0 \leq w_i \leq u$$

Two properties are worth keeping in mind. Over an infinite horizon Kelly beats any other fixed-fraction strategy almost surely. Over finite horizons it is aggressive: log utility implies a relative risk aversion of exactly 1, well below what most investors actually have.

We solve the empirical version by maximizing the sample mean of $\ln(1 + w^T \bar{r})$ over the training returns directly. Given the kurtosis in Table 1, using the Gaussian approximation would throw away precisely the tail behavior that log utility cares about.

2.2 Is the capped problem feasible?

Yes, and the argument is arithmetic rather than numerical. With a 20% cap, $1 \div 0.20 = 5$, so at least five positions must carry weight. The universe has ten assets, so feasible points always exist (any five-name equal-weight combination works). The feasible set is a non-empty compact convex polytope, and $\ln(1 + w^T r)$ is concave in w , so a maximum exists and is attained.

Table 2: Kelly portfolio weights

Ticker	Kelly (no cap)	Kelly (20% cap)	Status
TSLA	15.10%	20.00%	At cap
WMT	0.00%	20.00%	At cap
BAC	0.00%	0.00%	—
GS	39.06%	20.00%	At cap
LLY	0.00%	0.00%	—
MRK	0.00%	0.00%	—
GOOG	0.00%	20.00%	At cap
META	45.83%	20.00%	At cap
AAPL	0.00%	0.00%	—
XOM	0.00%	0.00%	—
Total	100.00%	100.00%	

The uncapped Kelly portfolio holds three names; the capped version holds exactly five, all at the ceiling. Growth rate falls from 57.98% to 53.67% a year once the 20% cap is imposed, a cost of 4.31 percentage points.

The uncapped solution holds three stocks out of ten. Meta at 45.8%, Goldman Sachs at 39.1%, and Tesla at 15.1% take the entire portfolio. Seven names get nothing, including Walmart, which has the second-highest return-to-volatility ratio in the universe.

This is what log utility does when it trusts its inputs. Kelly is more aggressive than mean-variance because it is chasing geometric growth, and it treats estimated means as if they were known. The three winners had the strongest realized growth over the training window, and the optimizer concentrated accordingly.

The capped solution is a pure corner. All five positions sit exactly at 20%. Nothing is interior. This tells us the cap, not the covariance structure, is deciding the composition so the optimizer would go further if allowed, and the constraint simply stops it at the boundary. That observation becomes important in Section 6, where denoising the covariance matrix turns out to change nothing at all.

3. Step 2: Kelly, Half-Kelly, and Double-Kelly

Fractional Kelly scales the allocation by a factor f and parks the remainder at the risk-free rate:

$$r_{p,2}(f) = f \cdot (w^T r_{p,2}) + (1 - f) \cdot r_f$$

$f = 0.5$ is half-Kelly, $f = 1$ is full Kelly, and $f = 2$ is double-Kelly, which borrows at the risk-free rate an amount equal to the portfolio value.

Table 3: Out-of-sample performance, 1 October to 31 December 2025

Measure	Half-Kelly ($f=0.5$)	Kelly ($f=1$)	Double-Kelly ($f=2$)
Cumulative return	4.66%	8.07%	14.42%
Ann. growth rate	19.64%	35.72%	69.96%
Sharpe ratio	1.4562	1.4562	1.4562
Ann. volatility	9.61%	19.21%	38.42%
Max drawdown	-3.16%	-6.52%	-13.07%
95% VaR (daily)	-0.91%	-1.84%	-3.70%
95% cVaR (daily)	-1.23%	-2.47%	-4.96%

Based on the 20%-capped Kelly weights. The three Sharpe ratios are identical to four decimal places, as shown in the discussion below.

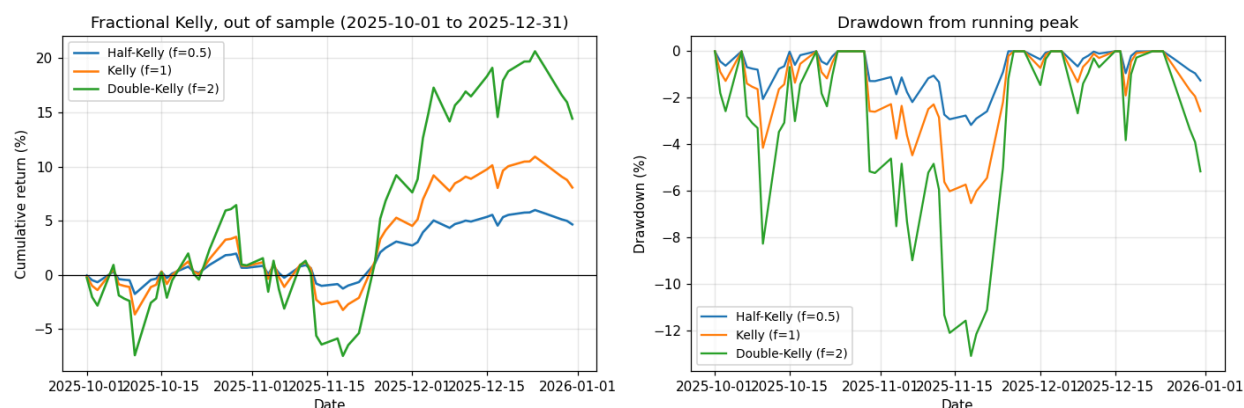


Figure 1: Fractional Kelly out-of-sample. Left: cumulative return. Right: drawdown from running peak. The three paths are the same shape scaled by f ; only the vertical axis changes.

3.1 Why the Sharpe ratios are identical

This is not a computational error, and it is the most useful thing in Table 3. Excess return scales exactly with the Kelly fraction:

$$r_{\text{excess}}(f) - r_f = f \cdot (w^T r_{\text{excess}} - r_f)$$

Both the mean and the standard deviation of excess return therefore scale by f , so the ratio between them is unchanged. **A Sharpe ratio cannot distinguish between Kelly fractions.** Any leverage decision judged solely on Sharpe alone is being judged on a statistic that is blind to it.

What separates them is in the rest of the table. Doubling the fraction doubled the volatility from 9.6% to 38.4%, doubled the drawdown from -3.2% to -13.1%, and quadrupled the cVaR loss from -1.2% to -5.0%. The investor's actual experience differs enormously across the three; the Sharpe ratio records none of it.

3.2 Did double-Kelly win?

Over this quarter, yes, with 14.42% against 8.07%. But the quarter was strongly positive, and leverage always wins in a rising market. The cross-validation in the next section gives a fairer test, and there the answer flips: in the weakest of the five folds, double-Kelly grew at 6.12% a year against full Kelly's 7.38%.

That reversal is the whole point of the Kelly criterion. Growth rate is not monotonic in leverage. Past the optimal fraction, the drag from compounding losses outweighs the gain from compounding wins, and the growth rate falls. Fold 3 is that regime showing up in our data. Double-Kelly took roughly twice the drawdown for a lower growth rate, being the worst possible trade.

4. Step 3: Backtesting with K-Fold Cross-Validation

4.1 Why plain K-fold fails on returns

Standard K-fold shuffles observations into folds, trains on K-1, and tests on the rest. Applied to a return series that is invalid for two reasons.

- **Leakage through time.** Shuffling puts future observations in the training set. A portfolio fitted on data that includes the test period will look better than it can possibly be in real time.
- **Leakage through serial correlation.** Even with contiguous folds, observations on either side of a boundary are linked to the test data through volatility clustering. A model trained on day t-1 knows something about day t.

4.2 Our procedure

Following López de Prado (2018), we purge and embargo. Purging removes training observations that overlap the test window in information terms; the embargo removes a further block immediately after the test fold, because serial correlation runs forward as well as backward.

1. Split all 501 observations into five contiguous folds in chronological order and no shuffling.
2. Treat each fold in turn as the test block, the remainder as training.
3. Purge and embargo five trading days on each side of the test block.
4. Re-solve the Kelly weights on the purged training data.
5. Evaluate on the untouched test fold for all three Kelly fractions.

The dispersion of results across folds is the measurement we care about. It estimates how much the Kelly weights depend on the particular period used to fit them.

Table 4: Cross-validated performance by Kelly fraction

Fraction	Growth (mean)	Growth (sd)	Growth (min)	Sharpe (mean)	Sharpe (sd)	Max DD (mean)	cVaR (mean)
0.5	18.44%	7.30%	6.44%	1.3427	0.7061	-5.31%	-1.38%
1.0	33.25%	15.79%	7.38%	1.3427	0.7061	-10.59%	-2.79%
2.0	64.82%	36.36%	6.12%	1.3427	0.7061	-20.47%	-5.59%

Five purged folds over the full two years. Note the minimum column: double-Kelly's worst fold (6.12%) is below full Kelly's worst (7.38%), despite double the average drawdown.

Table 5: Kelly weights re-estimated on each fold

Ticker	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Std. dev.
TSLA	20.0%	0.0%	0.0%	2.1%	20.0%	0.1061
WMT	20.0%	20.0%	20.0%	20.0%	20.0%	0.0000
BAC	19.4%	0.0%	0.0%	0.0%	14.4%	0.0944
GS	20.0%	20.0%	20.0%	20.0%	20.0%	0.0000
LLY	0.0%	20.0%	20.0%	20.0%	1.1%	0.1065
MRK	0.0%	0.0%	0.0%	0.0%	0.0%	0.0000

Ticker	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Std. dev.
GOOG	20.0%	20.0%	20.0%	20.0%	4.4%	0.0696
META	0.6%	20.0%	20.0%	17.9%	20.0%	0.0850
AAPL	0.0%	0.0%	0.0%	0.0%	0.0%	0.0000
XOM	0.0%	0.0%	0.0%	0.0%	0.0%	0.0000

Weights were re-estimated independently on each fold's training data. Five of ten names are stable; the other five swing across the full range from 0% to the 20% ceiling.

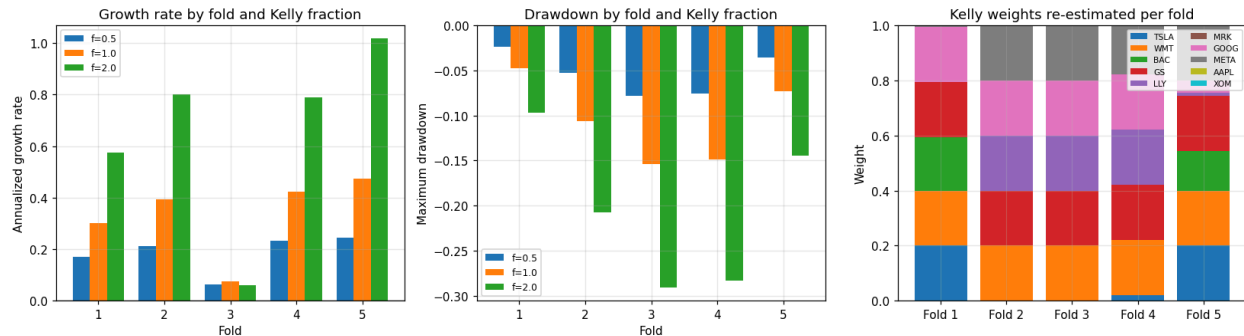


Figure 2: Cross-validation results. Left: growth rate by fold and Kelly fraction. Center: maximum drawdown. Right: Kelly weights are re-estimated per fold. Fold 3 is visibly the weak period across all three panels.

4.3 What the folds reveal

The weights are unstable, and Table 5 shows exactly where. Walmart and Goldman Sachs hold 20% in every fold. Merck, Apple, and Exxon are excluded in every fold. Those five are a genuine signal, and the data agrees about them no matter which period it looks at.

The other five are noise. Tesla is at the 20% ceiling in folds 1 and 5 and at zero in folds 2 and 3. Eli Lilly is the mirror image: zero in fold 1, at the ceiling in folds 2 through 4. These names swing across the entire feasible range depending on which stretch of history the optimizer happens to see. Their standard deviations across folds, around 0.10, are half the size of the cap itself.

The performance dispersion is severe. The mean Sharpe across folds is 1.34 with a standard deviation of 0.71, so the dispersion is more than half the mean. Fold 3 returned a Sharpe of 0.24, while fold 5 returned 2.10. Any single-period backtest of this strategy could report almost any number in that range, and the number would tell you more about the period chosen than about the strategy.

This is exactly what the exercise is meant to expose. A single out-of-sample quarter is not evidence, and we treat the Step 6 results accordingly.

5. Step 4: Denoising, Clustering, and De-toning

5.1 Denoising

The problem is straightforward. A covariance matrix on N assets from T observations has $N(N+1)/2$ parameters from NT data points. When T/N is small, the sample eigenvalues spread out far more than the true ones, and most of that spread is pure noise. The damage shows up at inversion: an optimizer inverting Σ divides by the smallest eigenvalues, which are the ones most corrupted. Small spurious eigenvalues become large spurious weights.

The method follows Marchenko and Pastur (1967). They give the eigenvalue distribution of a purely random correlation matrix. For $q = T/N$ and noise variance σ^2 , random eigenvalues fall between

$$\lambda_{\pm} = \sigma^2 (1 \pm \sqrt{1/q})^2$$

Anything above λ_+ is treated as a signal. We fit σ^2 by minimizing the distance between the empirical eigenvalue density and the theoretical Marchenko-Pastur density, then replace every eigenvalue below λ_+ with their common average. This preserves the trace, so total variance is unchanged, and the matrix is rebuilt from the modified spectrum.

5.2 Clustering: Hierarchical Risk Parity

The problem. Optimizers invert the covariance matrix, and inversion needs the matrix to be well conditioned. Correlated assets make it ill-conditioned, and the optimizer responds by taking large offsetting positions in near-substitutes, which is precisely the behavior that fails out-of-sample. López de Prado (2016) calls this Markowitz's curse where by the more diversification is needed, the less reliable the optimizer becomes.

The method. HRP never inverts anything, though it converts correlations to distances, $d_{ij} = \sqrt{((1-\rho_{ij})/2)}$, builds a hierarchical tree, reorders the covariance matrix so similar assets sit adjacent, then splits recursively by allocating between each pair of sub-clusters in inverse proportion to their variance.

The practical benefit is that weights depend on the structure of the correlation matrix rather than its inverse, so the result is stable and works even when the matrix is singular.

5.3 De-toning

The problem. In equity data, the largest eigenvector is the market factor, and it dominates every pairwise correlation. A clustering algorithm looking at raw correlations mostly sees the market, not the structure underneath.

The method. After denoising, we subtract the first eigenvector, then rescale back to a correlation matrix, and therefore, what remains describes how assets move relative to the market rather than with it.

One important caution is that detoning makes the matrix singular by construction. Removing one eigenvector leaves rank $N-1$, so the smallest eigenvalue is exactly zero. A detoned matrix must never be handed to an optimizer that inverts. It is safe for HRP, which does not invert.

5.4 Results on our data

Table 6: Denoising and detoning diagnostics

Quantity	Value	Reading
$q = T / N$	43.7	437 observations, 10 assets
Fitted noise variance σ^2	0.8611	From the MP density fit
MP cutoff λ_-	1.1413	Above this is signal
Eigenvalues retained	3 of 10	Seven judged to be noise
Largest eigenvalue	3.5508	The market factor
Smallest raw eigenvalue	0.2093	Most noise-corrupted
Denoised residual eigenvalue	0.5786	Common replacement value
Condition number, raw	16.96	—
Condition number, denoised	6.97	58.9% reduction
Condition number, detoned	3.72×10^{15}	Singular — do not invert
Market share of variance	34.7%	First eigenvector

The trace is preserved exactly at 10.0000 before and after denoising, which confirms that total variance is unchanged.

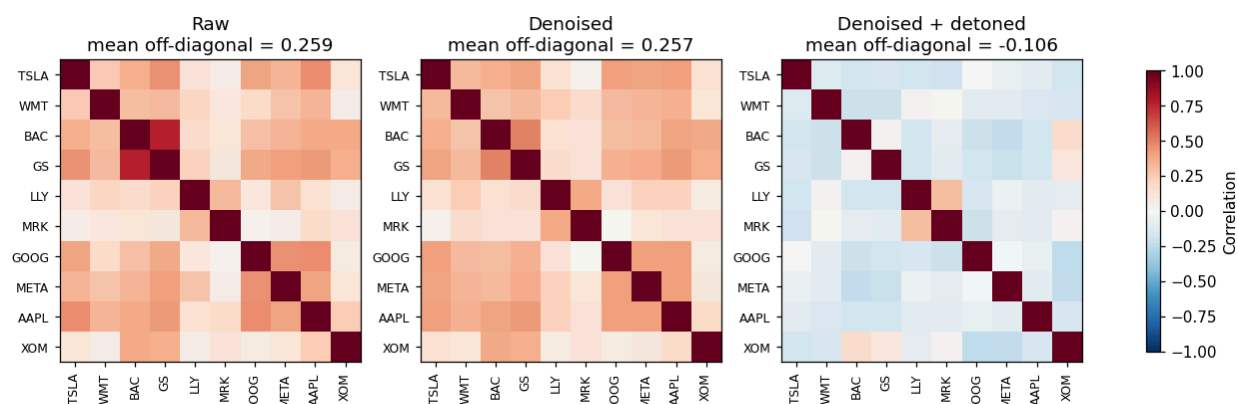


Figure 3 shows the correlation matrices under the three treatments. Denoising leaves the mean off-diagonal correlation almost unchanged at 0.257 against 0.259, while detoning drops it to -0.106 by removing the market factor.

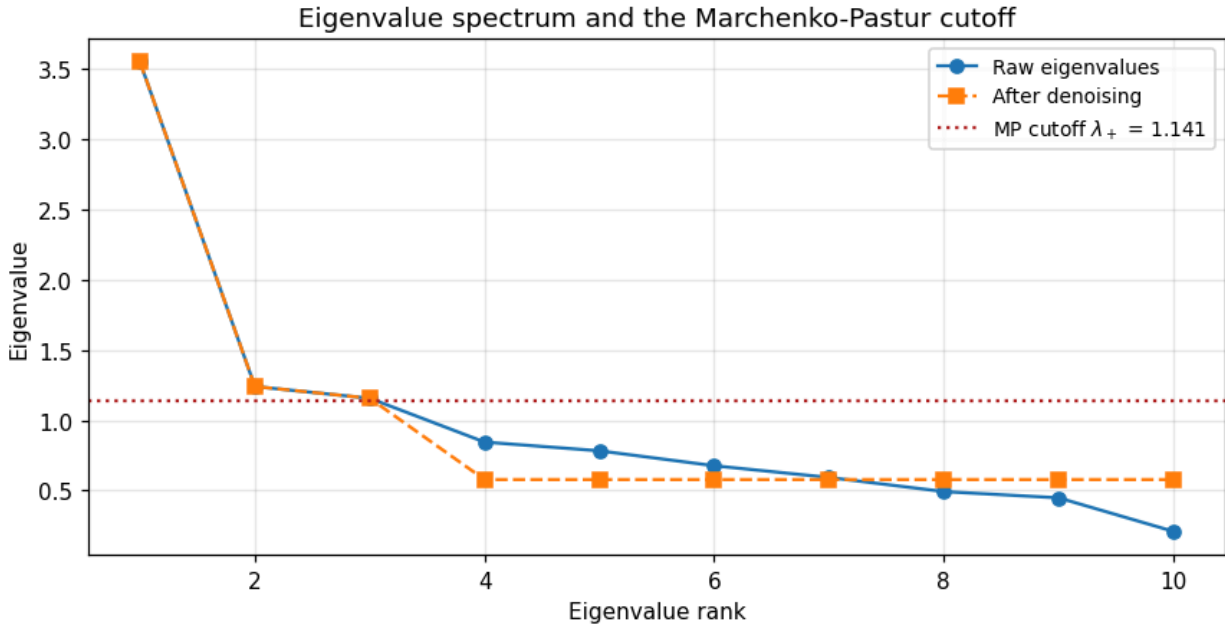


Figure 4 plots the eigenvalue spectrum against the Marchenko-Pastur cutoff. Three eigenvalues are clear $\lambda_+ = 1.1413$; the remaining seven are replaced by their common average of 0.5786.

Denoising worked as designed. The condition number fell from 16.96 to 6.97, a 58.9% improvement, and the smallest eigenvalue was lifted 2.8-fold from 0.209 to 0.579. Three of ten eigenvalues cleared the cutoff, so the procedure judged seven directions of the covariance matrix to be indistinguishable from noise.

Detoning did what the theory says and nothing more. Mean off-diagonal correlation fell from 0.257 to -0.106, so the market factor was genuinely the dominant shared component. The smallest eigenvalue came out at 4.7×10^{-16} , zero to machine precision, and the condition number rose to 3.7×10^{15} . We verified this numerically rather than asserting it, and it is why every detoned portfolio in this report is built with HRP.

Table 7: HRP weights under three covariance treatments

Ticker	HRP (raw)	HRP (denoised)	HRP (den. + detoned)
TSLA	2.23%	1.81%	2.35%
WMT	20.18%	20.90%	16.83%
BAC	6.55%	7.58%	12.82%
GS	4.99%	5.77%	8.06%
LLY	6.04%	6.85%	4.64%
MRK	12.43%	14.09%	10.64%
GOOG	10.69%	10.06%	11.23%
META	5.24%	5.48%	7.04%
AAPL	12.21%	8.59%	9.16%

Ticker	HRP (raw)	HRP (denoised)	HRP (den. + detoned)
XOM	19.44%	18.86%	17.24%
Effective N	7.47	7.48	8.27

Effective N is 1 divided by the sum of squared weights. All three HRP portfolios hold all ten names, against five for Kelly.

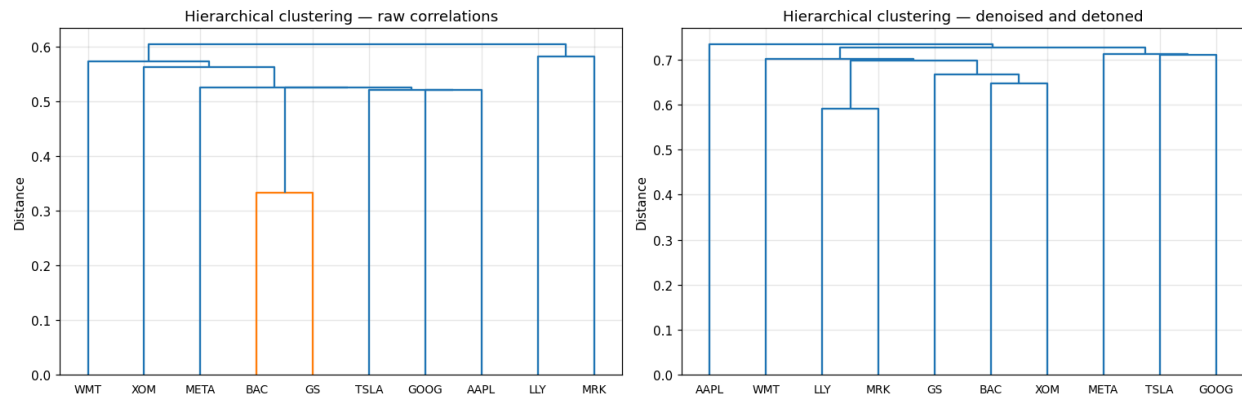


Figure 5: Hierarchical clustering dendrograms. Left: raw correlations. Right: denoised and detoned. Removing the market factor changes the cluster ordering substantially; the banks (BAC, GS) pair more tightly once market co-movement is stripped out.

Detoning changed the clustering, which is exactly what it is meant to do. The raw ordering starts with WMT, XOM, and META; the detoned ordering starts with AAPL, WMT, and LLY. More usefully, the effective number of positions rose from 7.47 to 8.27, so the detoned portfolio is genuinely better spread. And, with the market factor removed, the algorithm can see relative structure it previously could not.

One honest problem remains. HRP raw and HRP denoised put 20.18% and 20.90% into Walmart, breaching the 20% cap imposed in Step 1. HRP takes no constraints, but it allocates by hierarchy, and nothing in the algorithm knows about a position limit. Only the detoned version comes in under the cap, at 17.24%, and that is coincidence rather than design. A live mandate with a hard 20% limit would need the HRP output post-processed, and we flag the issue rather than quietly rescaling the numbers.

6. Steps 5 and 6: Which Combination Wins?

6.1 How the improvements can be sequenced

The three tools act on different objects, so they cannot be stacked in any order we like:

- **Denoising** modifies the covariance matrix and helps anything that inverts it.
- **Detoning** modifies it further and prevents inversion entirely.
- **HRP** replaces the optimizer and never inverts.

So denoising must therefore come before detoning, as detoning removes the leading eigenvector, and denoising needs the full spectrum to fit the cutoff. Detoned matrices can only be paired with HRP. To give denoising any chance to act on a Kelly allocation at all, we switch to the second-order approximation $g(w) \approx w^T \mu - \frac{1}{2} w^T \Sigma w$, because the empirical Kelly of Step 1 has no covariance matrix for these tools to improve.

Equal weight is included as a benchmark. DeMiguel, Garlappi, and Uppal (2009) found 1/N beats most optimized allocations out of sample, so any improvement that cannot beat it has not justified its complexity.

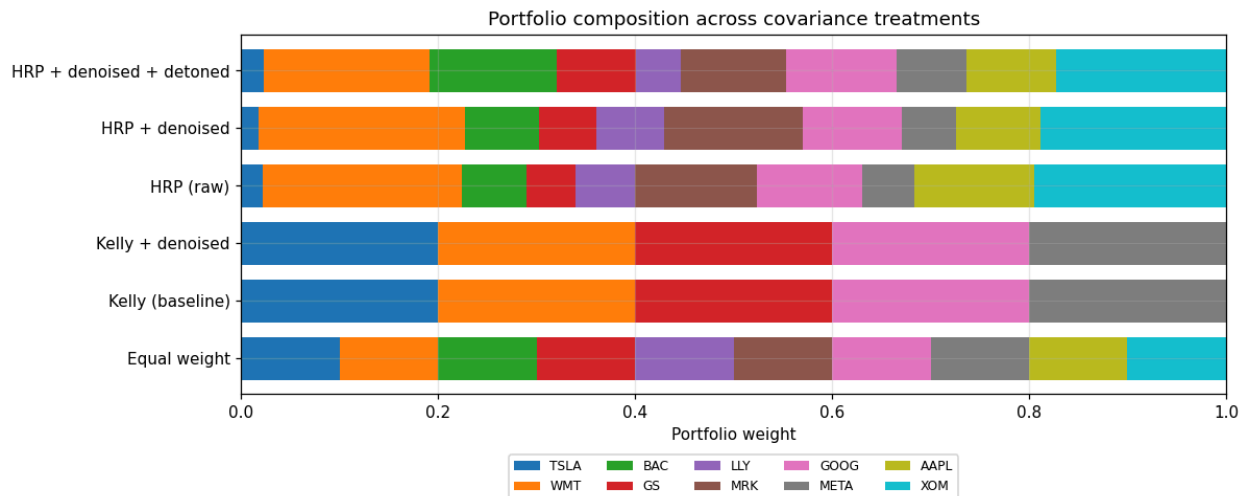


Figure 6 shows portfolio composition across treatments. The two Kelly portfolios hold five names at 20% each; the three HRP portfolios hold all ten.

6.2 Out-of-sample results

Table 8: Out-of-sample results, 1 October to 31 December 2025

Measure	Equal wt.	Kelly	Kelly + den.	HRP raw	HRP + den.	HRP+den+det
Cumulative return	12.89%	8.07%	8.07%	13.50%	13.97%	12.56%
Ann. growth rate	61.19%	35.72%	35.72%	64.63%	67.35%	59.32%
Sharpe ratio	3.3215	1.4562	1.4562	4.3115	4.4216	3.8715
Sortino ratio	5.2283	2.2336	2.2336	6.8547	7.0584	6.0215
Ann. volatility	13.32%	19.21%	19.21%	10.68%	10.79%	11.06%
Max drawdown	-3.61%	-6.52%	-6.52%	-3.07%	-2.96%	-3.17%
95% cVaR (daily)	-1.51%	-2.47%	-2.47%	-1.26%	-1.26%	-1.31%

Six metrics across all six portfolios. Note that Kelly and Kelly + denoised are identical in every column.

Table 9: Average rank across the six metrics (1 = best)

Portfolio	Cum. ret.	Growth	Sharpe	Sortino	Max DD	cVaR	Avg rank
HRP + denoised	1	1	1	1	1	2	1.17

Portfolio	Cum. ret.	Growth	Sharpe	Sortino	Max DD	cVaR	Avg rank
HRP (raw)	2	2	2	2	2	1	1.83
HRP + den. + detoned	4	4	3	3	3	3	3.33
Equal weight	3	3	4	4	4	4	3.67
Kelly + denoised	5	5.5	5	5	5.5	6	5.33
Kelly (baseline)	6	5.5	6	6	5.5	5	5.67

HRP + denoised ranks first on five of six metrics on the test quarter.

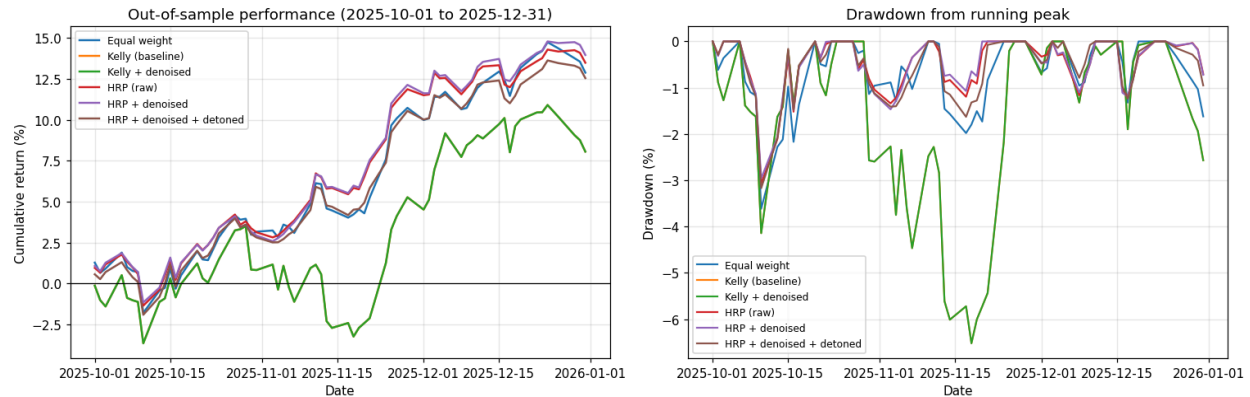


Figure 7: Out-of-sample performance and drawdown across all six portfolios. The three HRP variants cluster tightly together and above the two Kelly variants.

6.3 Question 1: What are the differences?

On the test quarter, the ordering is clear. HRP + denoised leads on five of six metrics with an average rank of 1.17, followed by HRP raw at 1.83. Both Kelly portfolios finish last.

The gap is large: HRP + denoised returned 13.97% against Kelly's 8.07%, at 10.79% volatility against 19.21%.

Two results deserve attention before anyone celebrates.

Denosing did nothing to the Kelly portfolio. Kelly and Kelly + denoised are identical to four decimal places in every column of Table 8, because their weights are identical. The reason is in Section 2.3: the capped Kelly solution is a pure corner, with all five positions at exactly 20%. The constraint fully determines the answer. Improving the covariance matrix cannot move a solution that is already pinned to the boundary of the feasible set.

Denosing barely moved HRP either. It improved the average rank from 1.83 to 1.17, but the underlying numbers moved by fractions of a percentage point, showing 13.97% against 13.50% cumulative return. The ranking overstates a difference that is small in economic terms.

6.4 The cross-validated check

A single 64-day quarter cannot establish that one method beats another. We therefore re-ran the whole comparison across the five purged folds, rebuilding every covariance matrix and every set of weights from each fold's training data.

Table 10: Cross-validated results across five purged folds

Portfolio	Growth (mean)	Growth (sd)	Sharpe (mean)	Sharpe (sd)	Max DD	cVaR
Equal weight	39.00%	27.97%	1.9936	1.7344	-9.70%	-2.43%
HRP (raw)	34.15%	28.63%	2.3182	2.3141	-7.49%	-1.92%
HRP + denoised	33.24%	27.31%	2.2144	2.2042	-7.66%	-1.95%
HRP + den. + detoned	35.58%	25.46%	2.2317	2.0079	-8.31%	-2.13%
Kelly (baseline)	33.25%	15.79%	1.3427	0.7061	-10.59%	-2.79%
Kelly + denoised	32.16%	14.76%	1.3049	0.6611	-10.67%	-2.79%

Every portfolio is rebuilt independently on each fold's training data. Standard deviations are reported because they are as informative as the means.

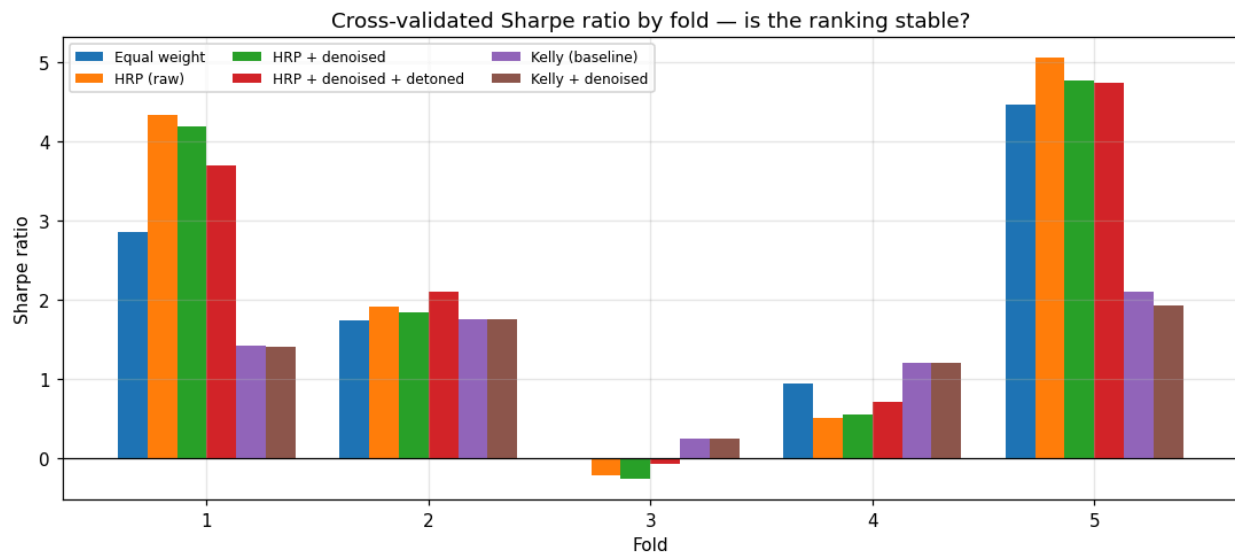


Figure 8: Cross-validated Sharpe ratio by fold. Fold 3 is negative for four of six portfolios. The dispersion across folds swamps the differences between portfolios.

The ranking does not survive. Equal weight posts the highest mean growth rate at 39.00%, ahead of every optimized alternative, including the HRP variants that won the test quarter. HRP keeps its edge on the risk metrics with a mean Sharpe of 2.32 against 1.99 and a mean drawdown of -7.49% against -9.70%, but its growth advantage disappears.

The dispersion is the real finding. HRP raw has a mean cross-validated Sharpe of 2.32 with a standard deviation of 2.31. The standard deviation equals the mean. In fold 3, four of the six portfolios posted negative Sharpe ratios. Against variation of that size, a difference of 0.1 in mean Sharpe is not a result, but it is noise with a decimal point.

What does hold across both tests is narrower but more credible: **HRP consistently reduces risk**. It beat every alternative on drawdown and cVaR on the test quarter and again on average across folds. That is the claim our data supports. The growth advantage is not.

6.5 Question 2: Why these differences?

$q = 43.7$ is the main explanation for the weak denoising result. We have 437 observations for 10 assets. The covariance matrix was already well estimated, so there was little noise available to remove. Denoising is built for the case where N approaches T : hundreds of assets in a short window. Our problem is nowhere near that regime, and the result is honest evidence about when the technique is worth applying rather than a failure of the technique.

The 20% cap already does part of HRP's job. A cap that forces at least five positions is itself a diversification constraint, and Jagannathan and Ma (2003) show weight limits act as shrinkage on the covariance estimate. Two mechanisms addressing the same problem do not add up. HRP's advantage over Kelly comes largely from holding ten names instead of five, and part of that advantage was already available through the constraint.

The corner solution blocks covariance improvements entirely. This is the sharpest lesson in the project. When a constraint binds on every active position, the optimizer's answer is determined by the feasible set rather than by the inputs. Any amount of work on the covariance matrix is wasted. Before investing in better estimation, check whether the constraint is deciding the outcome.

One quarter is one regime. Techniques that reduce estimation error pay off across regime changes. Our test window was a strong rising quarter for US large caps, and none of the folds contained a genuine regime break. The most favorable conditions for these tools were not present.

6.6 Question 3: Does the gain justify the complexity?

Taking the three techniques separately and judging against both tables:

HRP: Yes. It ranked first or second on every metric on the test quarter and retained the best mean Sharpe, drawdown, and cVaR across folds. It needs no matrix inversion, no return forecast, and no position constraint to produce a diversified portfolio. It also handles a singular covariance matrix, which nothing else here does. The implementation is perhaps forty lines. For a risk-focused mandate, this is the clearest win in the project.

Denoising: not at this q . It changed the Kelly portfolio not at all and moved HRP by fractions of a percentage point. The condition number genuinely improved by 58.9%, so the machinery works, but with 437 observations on 10 assets, better conditioning had nothing to fix. We would apply it if the universe grew past 50 names or the estimation window shortened below a year. At $q = 43.7$, it is effort without measurable return.

Detoning: not on its own evidence here. It ranked third on the test quarter, behind both other HRP variants, and its mean growth advantage across folds is well inside the noise. It did raise the effective number of positions from 7.47 to 8.27, and it produced the only HRP portfolio that respected the 20%

cap. There is a case for it where the market factor genuinely swamps the clustering, but our data does not make that case, as the market explained 34.7% of the variance, which is meaningful but not overwhelming.

And equal weight deserves a mention. It had the highest cross-validated growth rate of any portfolio tested, including every ML-improved variant, with no estimation and no code. DeMiguel, Garlappi, and Uppal (2009) predicted exactly this. Any recommendation from this project has to clear that benchmark, and on growth alone none of them does.

7. Conclusion

The Kelly criterion concentrates, and the cap decides the outcome. The uncapped Kelly portfolio held three of ten names, while the capped version held exactly five, every one at the 20% ceiling. That corner solution cost 4.31 percentage points of annual growth, and it made the covariance matrix irrelevant to the answer, which is why denoising later changed nothing.

A Sharpe ratio cannot evaluate leverage. Half-, full-, and double-Kelly all returned a Sharpe of 1.4562, because excess return scales linearly with the Kelly fraction. What separated them was drawdown, -3.16% against -13.07%, and cVaR, -1.23% against -4.96%. In the weakest cross-validation fold, double-Kelly grew more slowly than full Kelly while taking twice the drawdown, over-betting reducing growth, exactly as Kelly's theory predicts.

Cross-validation changed the conclusion. On the test quarter, HRP + denoised won five of six metrics. Across five purged folds, equal weight had the highest mean growth rate, and HRP's Sharpe standard deviation of 2.31 equaled its mean of 2.32. Only the risk improvement survived both tests: HRP reduced mean drawdown from -10.59% to -7.49% and cVaR from -2.79% to -1.92%. That is the claim we are willing to make.

The practical takeaway is about diagnosis before treatment. Denoising is built for the case where the number of assets approaches the number of observations. At $q = 43.7$, there was little noise to remove, and no amount of eigenvalue clipping could improve a solution already pinned to a constraint boundary. The right question is not which technique is best in general, but whether the specific failure a technique addresses is present in the problem at hand. In ours, the binding constraint mattered more than the estimation error, and identifying that is worth more than applying every tool available.

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